

Exercise 1

C/C++

```
struct student {  
    int id;  
    char name[20];  
    float gpa;  
};  
  
struct student s1 = {101, "Alice", 3.75};
```

Which command will cause a compilation error?

1. `s1.gpa = 3.90;`
2. `s1.id = 102;`
3. `s1.name[0] = 'B';`
4. `s1.name = "Bob";`
5. `s1.gpa = s1.gpa + 0.1;`

Exercise 2

C/C++

```
struct book {  
    char title[20];  
    int pages;  
};  
  
struct book library[ ] = {{"Math", 150}, {"Physics", 200}};
```

1. Number of elements in array library: _____
2. `sizeof(struct book)`: _____ bytes
3. Total memory size used by array library: _____ bytes

4. Value of library[1].pages: _____
5. Value of library[0].title: _____

Exercise 3

1. `struct item { char code[10]; int quantity; float price; }
x;`

Memory space: _____ bytes

2. `struct item { char name[12]; double price; } x;`

Memory space: _____ bytes

3. `int arr[3][5];`

Memory space: _____ bytes

4. `struct item { char title[15]; int page; } x;`

Memory space: _____ bytes